

Newton School of Technology, Rishihood University

Machine Learning

Traffic Speed Forecasting: ML Baselines and Graph Learning

Project Report

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Project GitHub Repository

https://github.com/ArPriCode/intelligent-traffic-forecasting

Declaration

We declare that this report, titled *Traffic Speed Forecasting: ML Baselines and Graph Learning*, documents the work carried out by Team Quanta as part of the Machine Learning course at Newton School of Technology, Rishihood University.

The report describes the datasets, preprocessing procedures, forecasting models, evaluation protocol, experimental results, and limitations. Numerical results are based on the project outputs supplied for this study. Unverified claims are identified explicitly.

Team Quanta
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We also thank all members of Team Quanta for their contributions to data analysis, model development, experimentation, and documentation.

Abstract

Traffic-speed forecasting supports intelligent transportation systems by estimating future road conditions from historical observations. This project compares persistence, segment-wise multi-output Random Forest, pooled XGBoost, and a trained Graph Attention Network (GAT) using five-minute METR-LA traffic-speed observations.

The evaluation uses chronological training, validation, and test partitions of 70%, 10%, and 20%. Forecast horizons are 5, 15, 30, and 60 minutes. Missing and zero-speed observations are handled through causal input filling and masked target evaluation. Performance is assessed using Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and Root Mean Squared Error (RMSE).

Random Forest achieves the lowest overall MAE at all four horizons. The GAT incorporates spatial attention over a fixed graph of 207 sensors but does not outperform the tree-based models in the reported experiment. This demonstrates that graph structure alone does not guarantee better forecasting accuracy.

The project also examines peak and off-peak performance, graph construction, data quality, and experimental limitations. The supplied traffic CSVs contain speed observations only; flow and occupancy are not treated as measured variables.

Keywords: Traffic forecasting, METR-LA, Random Forest, XGBoost, Graph Attention Network, spatial graph, time-series prediction.

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1. Introduction

1.1 Background

Traffic congestion increases travel delays, fuel consumption, and transportation costs. Traffic-speed forecasting estimates future road conditions using historical observations and supports traffic monitoring, route planning, and transportation decisions.

Traffic measurements are spatially related. Conditions at one road segment may be associated with conditions at nearby segments. Conventional machine learning models learn temporal patterns through engineered features, while graph-based models represent sensors as nodes and their relationships as edges.

1.2 Problem Statement

This project compares conventional machine learning baselines with a Graph Attention Network for forecasting traffic speed at multiple future horizons using METR-LA.

The study investigates:

1. How accurately can persistence, Random Forest, and XGBoost forecast traffic speed?
2. Does graph attention improve accuracy over tree-based baselines?
3. How does error change with forecast horizon?
4. How do peak and off-peak periods affect performance?

1.3 Objectives

- Analyze traffic data and data-quality issues.
- Establish a persistence baseline.
- Train Random Forest and XGBoost models.
- Construct a spatial graph for METR-LA sensors.
- Train and evaluate a Graph Attention Network.
- Compare errors across horizons and traffic periods.
- Document limitations and future improvements.

1.4 Scope

METR-LA is the primary dataset for quantitative model comparison. PEMS-BAY is included in dataset documentation and exploratory analysis, but its model-comparison results are not reported here.

The project is not a deployed traffic-control system. Signal-control simulation is an optional experiment, separate from the forecasting benchmark.

2. Literature Review

2.1 Traffic-Speed Forecasting

Traffic forecasting uses statistical time-series models, conventional machine learning, and deep learning. Historical speed provides predictive information, although uncertainty generally increases with forecast horizon.

Persistence is a useful baseline because it assumes that the latest observed speed remains unchanged.

2.2 Tree-Based Models

Random Forest combines predictions from multiple decision trees and models nonlinear relationships among historical speed, time features, and sensor-specific behavior.

XGBoost uses gradient-boosted decision trees to learn complex relationships through sequential optimization. This project uses a pooled, sensor-aware XGBoost model.

2.3 Graph-Based Forecasting

A sensor network can be represented as a graph $G = (V, E)$, where V denotes sensors and E denotes their relationships. Graph models aggregate information from neighboring sensors.

Graph Attention Networks learn attention weights over graph neighborhoods. The architecture was introduced by Veličković et al. [?]. Diffusion Convolutional Recurrent Neural Networks provide another influential approach to traffic forecasting [?].

2.4 Research Motivation

This project investigates whether spatial attention improves forecasting accuracy over conventional baselines. Reporting negative results is important because a complex model should not automatically be considered superior when its held-out metrics are worse.

3. Dataset and Exploratory Analysis

3.1 Dataset Overview

The project examines two public traffic-speed datasets. Quantitative model comparison is performed on METR-LA.

Table 3.1: Dataset overview

Dataset	Time steps	Sensors	Date range	Missing/zero
METR-LA	34,272	207	2012-03-01 to 2012-06-27	8.11%
PEMS-BAY	52,116	325	2017-01-01 to 2017-06-30	0.0031%

The datasets contain five-minute speed observations. Missing and zero percentages are dataset-level summaries provided by the project analysis.

3.2 Data Quality and Exploratory Analysis

Zero speeds are treated as missing according to the project preprocessing assumptions. Forward filling is used only for causal inputs, while missing targets are excluded from training and evaluation.

The supplied CSVs contain speed only. Flow and occupancy are not available as measured variables.

Exploratory analysis covers:

- Speed time series and distributions.
- Hourly and weekday patterns.
- Sensor-level heatmaps and correlations.
- Missingness and zero-speed analysis.
- Sensor coordinates and road metadata.

3.3 METR-LA Sensor Graph

Sensor IDs, locations, and road distances are sourced from the DCRNN repository at commit `602afd9d767d3aa1c9b3eac51710d`

The official ID order aligns with all 207 traffic columns. A Gaussian distance kernel with threshold 0.1 constructs the adjacency matrix. The matrix is symmetrized, and self-loops are added for graph learning.

Table 3.2: METR-LA graph characteristics

Characteristic	Value
Nodes	207
Undirected links	1,313
Mean degree	Approximately 12.7
Graph type	Symmetrized distance graph
Self-loops	Added for model computation

The graph represents spatial proximity based on road distance. It does not encode verified upstream/downstream direction.

4. Preprocessing and Feature Engineering

4.1 Chronological Splitting

Observations are divided chronologically into training, validation, and test subsets.

Table 4.1: Chronological data partition

Partition	Proportion
Training	70%
Validation	10%
Testing	20%

Chronological splitting preserves time order and avoids randomly mixing future observations into training.

4.2 Forecast Horizons

Table 4.2: Forecast horizons

Horizon	Five-minute intervals
5 minutes	1
15 minutes	3
30 minutes	6
60 minutes	12

4.3 Input Processing

The pipeline preserves chronological order, treats zero-speed readings as missing, forward-fills only causal inputs, excludes missing targets, fits sensor-specific scaling using training timestamps, and constructs speed lags and temporal features.

The GAT uses six consecutive observations, representing 30 minutes of history. Tree models use causal lags and time features with no more than 25 minutes of input history under the specified evaluation protocol.

4.4 Temporal Features and Target Masking

Cyclic time-of-day features are represented as

$$x_{\text{hour},1} = \sin(\theta_t), \qquad x_{\text{hour},2} = \cos(\theta_t).$$

Equivalent cyclic features can represent weekday patterns.

For sensor i at time t, the target mask is

$$m_{i,t} = \begin{cases} 1, & \text{if the target is observed and nonzero,} \\ 0, & \text{otherwise.} \end{cases}$$

Only targets with $m_{i,t} = 1$ contribute to training losses and evaluation metrics. The GAT uses masked loss so unavailable targets do not affect optimization.

5. Methodology and System Architecture

5.1 Pipeline

The system loads traffic observations and sensor metadata, performs data-quality checks, creates chronological splits, generates causal features, trains four forecasting approaches, and evaluates them at all specified horizons.

5.2 Persistence Baseline

Persistence predicts future speed using the latest available observation:

$$\hat{y}_{t+h} = y_t.$$

It measures whether more complex models improve upon the assumption that current conditions remain unchanged.

5.3 Random Forest

A multi-output Random Forest is trained separately for each sensor. It uses causal speed lags and cyclic time features to predict four horizons. Training is capped at 2,000 timestamps per sensor.

5.4 Pooled XGBoost

Pooled XGBoost combines observations from multiple sensors using speed lags, cyclic time features, and sensor identity. Up to 50,000 observed training targets are used per horizon.

Pooling enables shared pattern learning while retaining sensor-specific information.

5.5 Graph Attention Network

The GAT learns representations from historical speeds and neighboring sensors in the fixed graph. Attention weights allow connected sensors to contribute differently.

The implementation uses six consecutive five-minute speed observations, cyclic temporal features, training-only per-sensor scaling, masked loss, and validation-based early stopping.

It was trained for six epochs on Apple Metal Performance Shaders (MPS). The best reported validation masked Mean Squared Error was 0.4220.

5.6 Comparison Protocol

All models use the same test origins and valid target counts at each horizon, but feature representations and training volumes differ. Random Forest is capped at 2,000 timestamps per sensor, XGBoost at 50,000 observed targets per horizon, and GAT uses eligible training windows.

The comparison is chronological but not equal-compute: training examples, model capacity, and computational costs are not matched.

6. Evaluation Metrics and Protocol

6.1 Evaluation Metrics

Mean Absolute Error (MAE) is

$$\text{MAE} = \frac{1}{N}\sum_{j=1}^{N}|y_j - \hat{y}_j|.$$

Mean Absolute Percentage Error (MAPE) is

$$\text{MAPE} = \frac{100}{N}\sum_{j=1}^{N}\left|\frac{y_j - \hat{y}_j}{y_j}\right|.$$

Root Mean Squared Error (RMSE) is

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{j=1}^{N}(y_j - \hat{y}_j)^2}.$$

Lower values indicate better performance. MAPE excludes zero and missing targets to avoid division by zero.

6.2 Peak and Off-Peak Periods

Peak periods are weekdays from 07:00–09:59 and 16:00–18:59. All other times, including weekends, are classified as off-peak.

Metrics are calculated over observed nonzero targets. Each model uses the same valid target count at each horizon.

7. Experimental Results

7.1 Overall Held-Out Results

Table 7.1 reports the held-out METR-LA metrics for the four models.

Table 7.1: Overall held-out METR-LA metrics

Horizon	Model	MAE	MAPE (%)	RMSE
5 min	Persistence	2.814	6.489	4.627
5 min	Random Forest	**2.519**	**6.327**	**4.281**
5 min	XGBoost	2.554	6.435	4.326
5 min	GAT	2.879	7.206	4.511
15 min	Persistence	3.499	8.612	6.418
15 min	Random Forest	**3.119**	**8.431**	**5.702**
15 min	XGBoost	3.239	8.846	5.873
15 min	GAT	3.466	9.168	5.830
30 min	Persistence	4.207	10.858	8.052
30 min	Random Forest	**3.635**	**10.401**	**6.706**
30 min	XGBoost	3.854	11.106	6.988
30 min	GAT	4.087	11.434	6.965
60 min	Persistence	5.381	14.535	10.345
60 min	Random Forest	**4.192**	**12.623**	**7.657**
60 min	XGBoost	4.542	13.696	7.982
60 min	GAT	4.898	14.328	8.227

Lower values are better. Bold values identify the lowest metric within each horizon.

7.2 Valid Target Counts

Table 7.2: Valid held-out target counts per model

Horizon	Valid targets
5 minutes	1,244,846
15 minutes	1,244,834
30 minutes	1,244,816
60 minutes	1,244,780

Valid counts decrease slightly at longer horizons because targets must be available at corresponding future times. The reported counts are identical across models within each horizon.

7.3 Model Comparison Figure

Add the model-comparison plot from `results/figures/27_metr_la_peak_offpeak_model_comparison.png`.

Figure 7.1: Peak and off-peak forecasting comparison.

Random Forest achieves the lowest overall MAE at every horizon. Its MAE increases from 2.519 at five minutes to 4.192 at 60 minutes, indicating increasing forecasting difficulty.

XGBoost improves upon persistence in overall MAE at every horizon but remains behind Random Forest. GAT has higher overall MAE than both tree-based models. It improves upon persistence at 15, 30, and 60 minutes, but not at five minutes. Graph attention therefore does not establish an accuracy advantage in this experiment.

7.4 Peak-Period MAE

Random Forest achieves the lowest peak-period MAE at every horizon. All models have higher peak-period errors than their off-peak errors.

7.5 Off-Peak MAE

Random Forest also achieves the lowest off-peak MAE at all four horizons. The difference between peak and off-peak errors shows that predictability varies across traffic periods.

Table 7.3: Peak-period MAE

Model	5 min	15 min	30 min	60 min
Persistence	3.287	4.565	5.831	7.804
Random Forest	**3.189**	**4.341**	**5.324**	**6.374**
XGBoost	3.243	4.535	5.808	7.277
GAT	3.601	4.737	6.001	7.940

Table 7.4: Off-peak MAE

Model	5 min	15 min	30 min	60 min
Persistence	2.695	3.230	3.797	4.770
Random Forest	**2.350**	**2.812**	**3.210**	**3.643**
XGBoost	2.381	2.913	3.362	3.853
GAT	2.696	3.146	3.605	4.131

7.6 Discussion

The principal findings are:

1. **Random Forest leads the comparison.** It achieves the lowest overall MAE and leads both peak and off-peak comparisons.
2. **Longer horizons are harder to predict.** All four models show increasing MAE between five and 60 minutes.
3. **Graph attention does not guarantee better accuracy.** The GAT uses a spatial graph and learned attention but does not outperform the tree-based models.

These findings support conventional tree-based models as strong baselines for the data, features, training budgets, and evaluation protocol used here.

8. Optional Signal-Control Simulation

8.1 Scope and Status

The project materials describe an optional intersection queuing simulation in `notebooks/09_signal_simulation.ipynb`. It compares a fixed-time signal policy with a forecast-aware adaptive policy and reports a 42.1% reduction in total intersection vehicle delay during peak congestion.

Qualification: this is a reported simulation result, not a measured real-world improvement. Retain the percentage only after checking the notebook, output figure, demand assumptions, delay calculations, and reproducibility.

8.2 Policies and Assumptions

The fixed-time policy uses a static 90-second cycle with a reported 53%/36% green-time split. The forecast-aware policy adjusts green splits according to anticipated queue buildup using a max-pressure approach.

The simulation references the Greenshields relationship:

$$q = kv,$$

where q is flow, k is density, and v is speed. Because the supplied CSVs contain speed only, inferred flow or density must be documented as a modeling assumption.

8.3 Interpretation

If reproduced, this simulation can demonstrate how forecasts might inform signal timing under specified assumptions. It does not establish control of actual road infrastructure or generalizable real-world delay reduction.

Add the verified output from
`results/figures/28_simulated_signal_policy_experiment.png`.

Figure 8.1: Optional signal-policy simulation. Include only after verifying the underlying results.

9. Limitations

9.1 Data Limitations

The supplied traffic CSVs contain speed observations only. Flow and occupancy are unavailable as measured variables. Treating zero speeds as missing may also exclude genuine stationary-traffic events.

9.2 Graph Limitations

The symmetrized road-distance graph represents spatial proximity but does not encode verified upstream/downstream direction or direct travel times.

9.3 Model Comparison Limitations

Feature representations and training volumes differ. Random Forest is capped at 2,000 timestamps per sensor, XGBoost at 50,000 observed targets per horizon, and GAT uses eligible training windows. Computational cost and model capacity are not equalized.

The GAT was trained for six epochs on MPS, with a best validation masked MSE of 0.4220. Additional tuning, training, or alternative graph construction may change its performance.

9.4 Signal-Control Limitations

The forecasting benchmark is not a live traffic-control system. Signal simulation requires independent verification.

Real-world evaluation would need calibrated demand, intersection geometry, phase sequences, detector data, and a suitable traffic simulator such as SUMO or CityFlow. The available results do not establish deployment readiness.

10. Conclusion and Future Work

10.1 Conclusion

This project evaluated persistence, segment-wise Random Forest, pooled XGBoost, and a Graph Attention Network on METR-LA at 5-, 15-, 30-, and 60-minute horizons.

Random Forest achieved the lowest overall MAE at all horizons: 2.519, 3.119, 3.635, and 4.192, respectively. It also led peak and off-peak MAE comparisons. XGBoost improved upon persistence in overall MAE but did not surpass Random Forest.

The GAT implemented spatial attention over a fixed 207-node sensor graph, but its overall MAE remained higher than that of both tree-based models. This negative result shows that graph-learning complexity did not translate into better accuracy in the reported experiment.

The project documents data quality, causal preprocessing, chronological evaluation, graph limitations, and differences in training budgets. Signal-control simulation remains exploratory and does not establish real-world performance.

10.2 Future Work

- Tune GAT depth, attention heads, hidden dimensions, dropout, learning rate, and training duration.
- Compare fixed-distance graphs with learned or directed sensor graphs.
- Run multiple seeds and report variability.
- Compare models under more closely matched budgets.
- Inspect per-sensor errors and additional metrics.
- Evaluate PEMS-BAY under the same protocol.
- Independently validate signal simulations using documented assumptions and calibrated scenarios.

Reproducibility and Project Deliverables

The project repository is available at:

https://github.com/ArPriCode/intelligent-traffic-forecasting

The project materials identify these notebooks:

- `notebooks/05_baselines.ipynb`: baseline models.
- `notebooks/06_xgboost.ipynb`: pooled XGBoost.
- `notebooks/07_backtest.ipynb`: chronological evaluation.
- `notebooks/08_gat_model.ipynb`: graph and GAT.
- `notebooks/09_signal_simulation.ipynb`: optional simulation.

Metrics, predictions, adjacency data, model state, and figures are described under `results/metrics/`, `results/predictions/`, and `results/figures/`. The detailed metrics file is identified as `results/metrics/metr_la_model_comparison.csv`.